

PROJECT REPORT

Analysing Mental Health in Student Ecosystem

A Tableau-based Data Analysis and Visualization Project

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Course	B.Com (Hons.)
Project Area	Data Analysis & Visualization using Tableau

Academic Project Report • 2026

Declaration

I, Rudr Arora, student of B.Com (Hons.) at IITM Janakpuri, declare that this project report titled “Analysing Mental Health in Student Ecosystem” has been prepared as part of my academic project work. The dashboard and analysis presented in this report were developed using Tableau for the purpose of understanding patterns in the given student dataset.

The report is intended for academic and analytical use. The observations shown here describe patterns in the dataset and should not be treated as medical diagnosis or professional mental-health advice.

Acknowledgement

I would like to thank my college and faculty for providing the opportunity to work on a practical data visualization project. Working on this topic helped me understand how raw data can be converted into visual information that is easier to interpret. I also gained practical experience in Tableau, including creating charts, applying measures and dimensions, and arranging individual worksheets into a dashboard.

1. Abstract

Mental health has become an important part of the student experience because academic pressure, sleep patterns, study routines and social factors can affect how students feel and perform. This project uses a student-level dataset to explore selected indicators related to mental health and academic life. The analysis was carried out in Tableau and presented through an interactive dashboard.

The dashboard contains 1,000 student records and brings together measures such as stress level, sleep, study hours, academic performance index, depression level, gender and heart rate variability observations. The purpose is to identify broad patterns and demonstrate how visualization can make the data easier to understand.

2. Introduction

Student life involves academic work, personal responsibilities, changing routines and social pressure. Because of this, mental well-being can be influenced by several factors at the same time. A dataset containing information about students can therefore be useful for finding patterns that may not be obvious when individual records are viewed one by one.

For this project, Tableau was used to convert the available data into a set of visualizations and combine them into a single dashboard. The dashboard was designed to give a quick overview first and allow the individual charts to be viewed in more detail when required.

3. Objectives of the Project

- To study selected mental-health-related indicators in a student dataset.
- To compare stress and depression-related measures across gender groups.
- To examine the distribution of students by mental health risk-related records.
- To observe the pattern between study hours and the academic performance index.
- To visualize sleep, stress, depression and heart-rate-variability-related information.
- To build a clear Tableau dashboard that presents the major findings in one place.
- To develop practical skills in data visualization, dashboard design and interpretation.

4. Scope of the Study

The scope of this project is limited to the variables available in the supplied student dataset. The analysis focuses on descriptive patterns rather than clinical assessment. The dashboard can help a viewer understand how the recorded indicators are distributed across the sample, but it cannot establish that one factor directly causes another.

5. Tools and Technologies Used

- Tableau – used for data visualization, worksheet creation and dashboard design.
- Student dataset – used as the source for the analysis.
- Microsoft Word – used for preparing the project report.
- Screenshots – used to document the Tableau worksheets and final dashboard.

6. Data and Methodology

The project followed a simple data-analysis workflow. The student dataset was used as the source for the Tableau workbook. Relevant fields were then used as dimensions or measures depending on the visualization. Aggregations such as count, average and sum were applied where appropriate.

- Load and inspect the student data in Tableau.
- Identify the variables useful for the project objective.
- Create individual worksheets for the required views.
- Use aggregation methods such as AVG, SUM and COUNT.
- Compare gender groups using charts and a treemap.
- Create a study-hours versus academic-performance visualization.
- Arrange the worksheets into a single dashboard.
- Review the dashboard and record the main observations.

7. Key Variables Used

Variable / Measure	Use in Dashboard	Visualization / Summary
Gender	Group comparison	Treemap, bar chart, pie chart
Stress Level	Overall and gender comparison	Average / treemap
Sleep	Overall sleep indicator	Dashboard KPI
Study Hours Per Day	Academic routine	X-axis of performance chart
Academic Performance Index	Academic outcome indicator	Y-axis of performance chart
Depression Level	Depression-related measure	Gender comparison / total
Mental Health Risk	Risk distribution	Gender-wise count
Heart Rate Variability	Physiological indicator	Gender-wise observation count

8. Tableau Dashboard

The final dashboard was designed to provide an overall snapshot of the student ecosystem. The top section highlights key figures, while the remaining sections provide visual comparisons of stress, mental health risk, heart rate variability, study hours and depression.

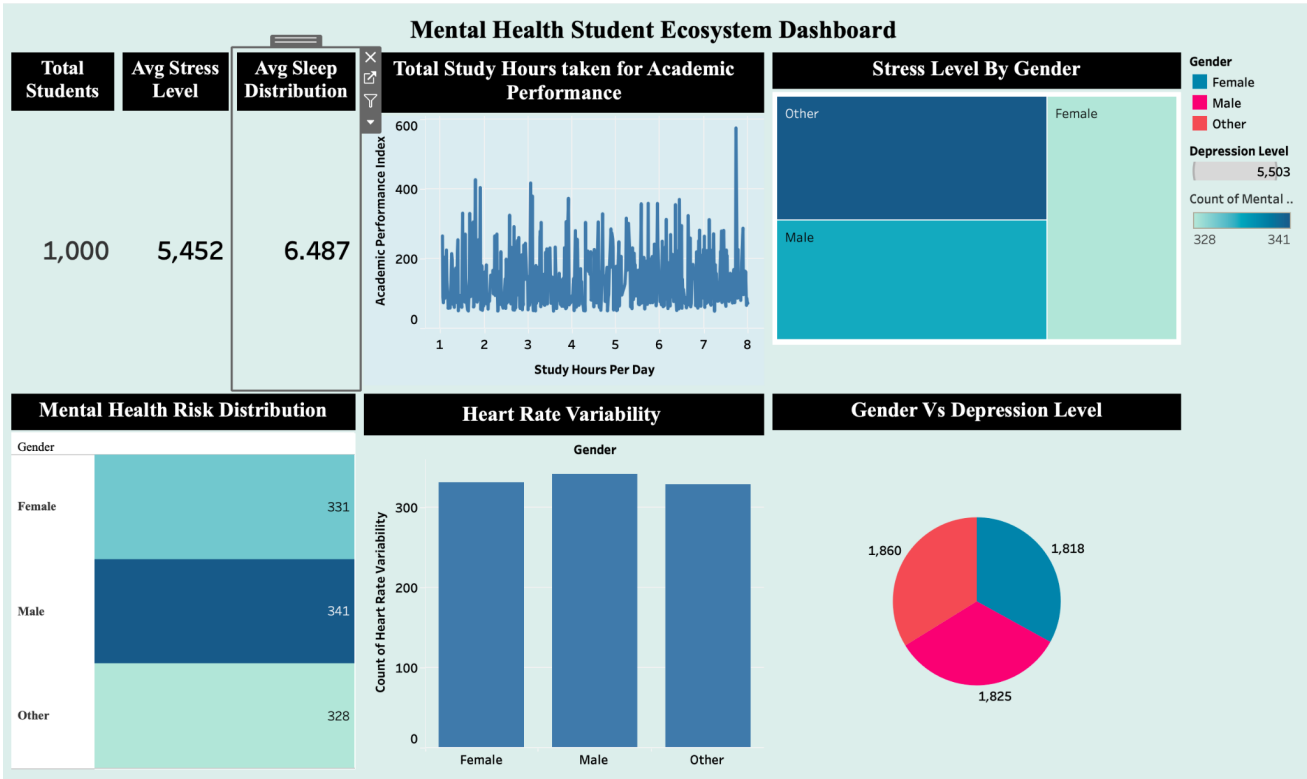


Figure 1: Final Tableau dashboard – Analysing Mental Health in Student Ecosystem

9. Individual Visualizations and Interpretation

9.1 Mental Health Risk Distribution

The mental health risk distribution view shows the number of records by gender. The displayed counts are 331 for Female, 341 for Male and 328 for Other. The values are close to one another, indicating a fairly balanced distribution across the three gender categories.

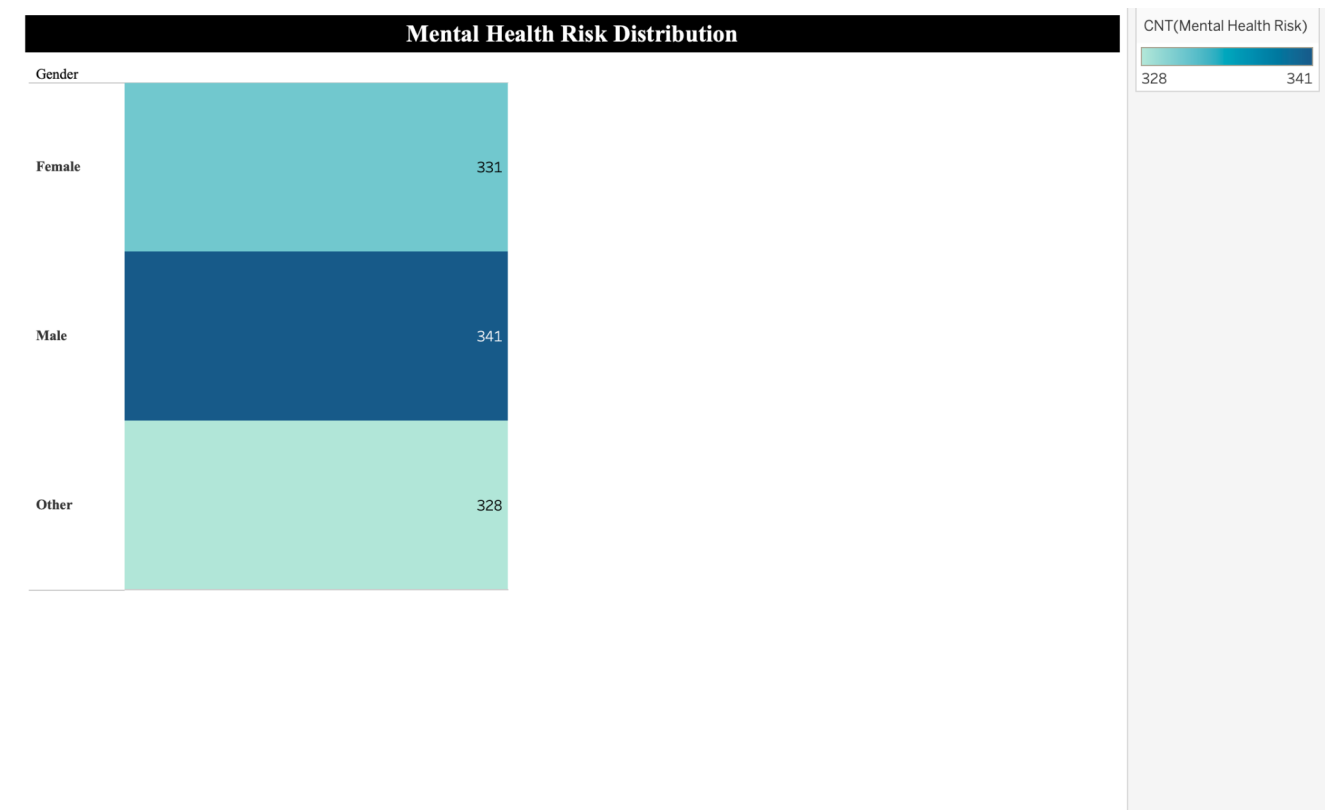


Figure 2: Mental Health Risk Distribution

9.2 Heart Rate Variability

The Heart Rate Variability worksheet compares the count of recorded observations across gender. The bars are close in height, with Male slightly higher than Female and Other. In this worksheet, the vertical axis is based on COUNT, so the chart represents the number of observations rather than an average HRV value.

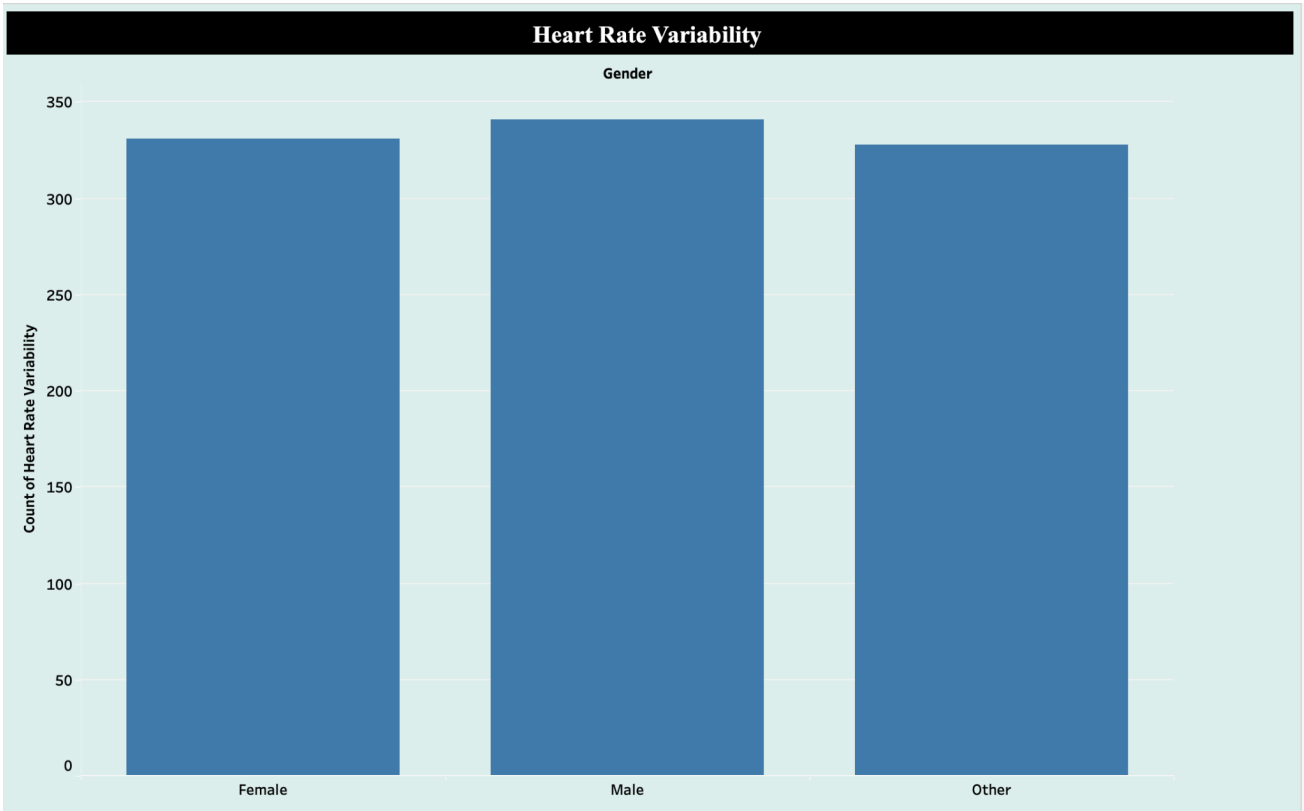


Figure 3: Heart Rate Variability by Gender

9.3 Stress Level by Gender

The treemap compares stress level across the three gender categories. The displayed average stress values range approximately from 5.30 to 5.61. The values are relatively close, so the visual should be interpreted as a small difference rather than a major gap.

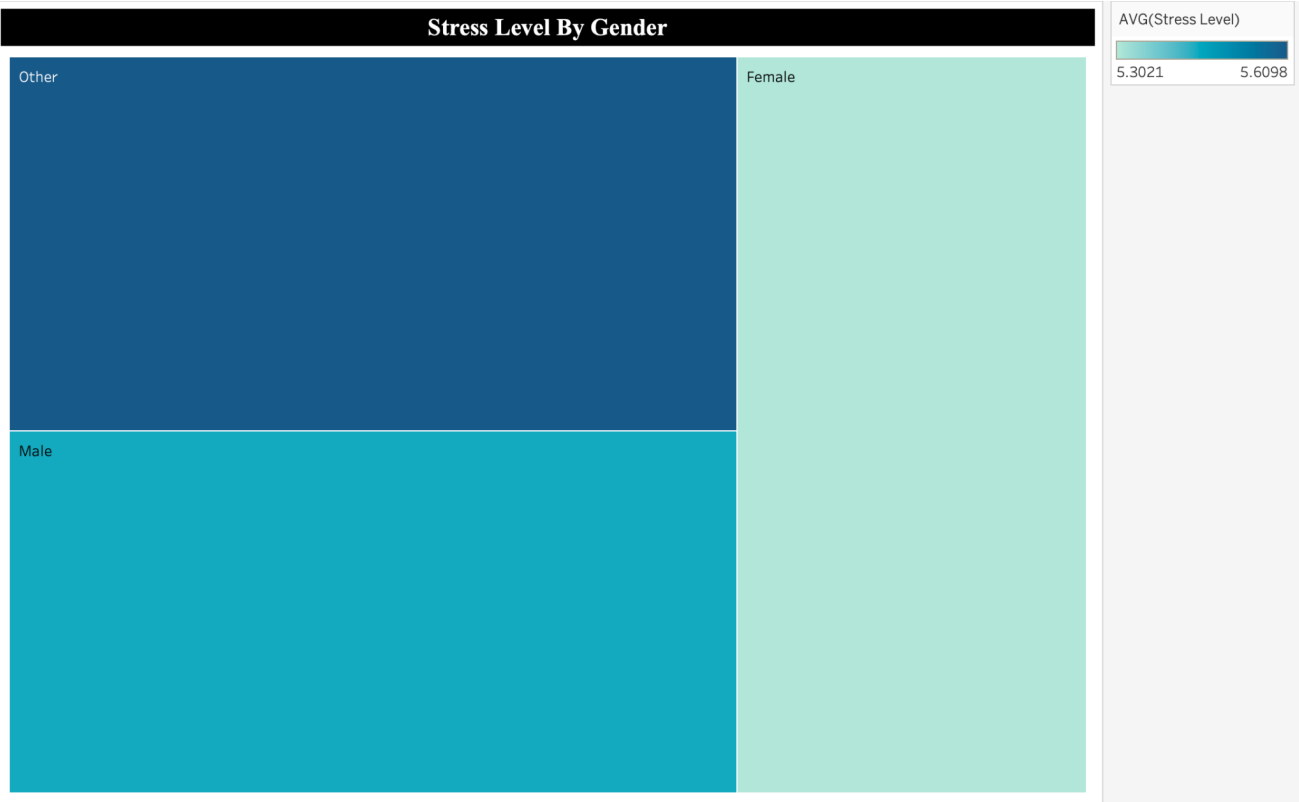


Figure 4: Stress Level by Gender

9.4 Study Hours and Academic Performance

This chart plots Study Hours Per Day against the Academic Performance Index. The visualization shows substantial variation in performance at different study-hour levels. Some high values appear at different study-hour levels, suggesting that study time alone does not explain all of the variation in academic performance.

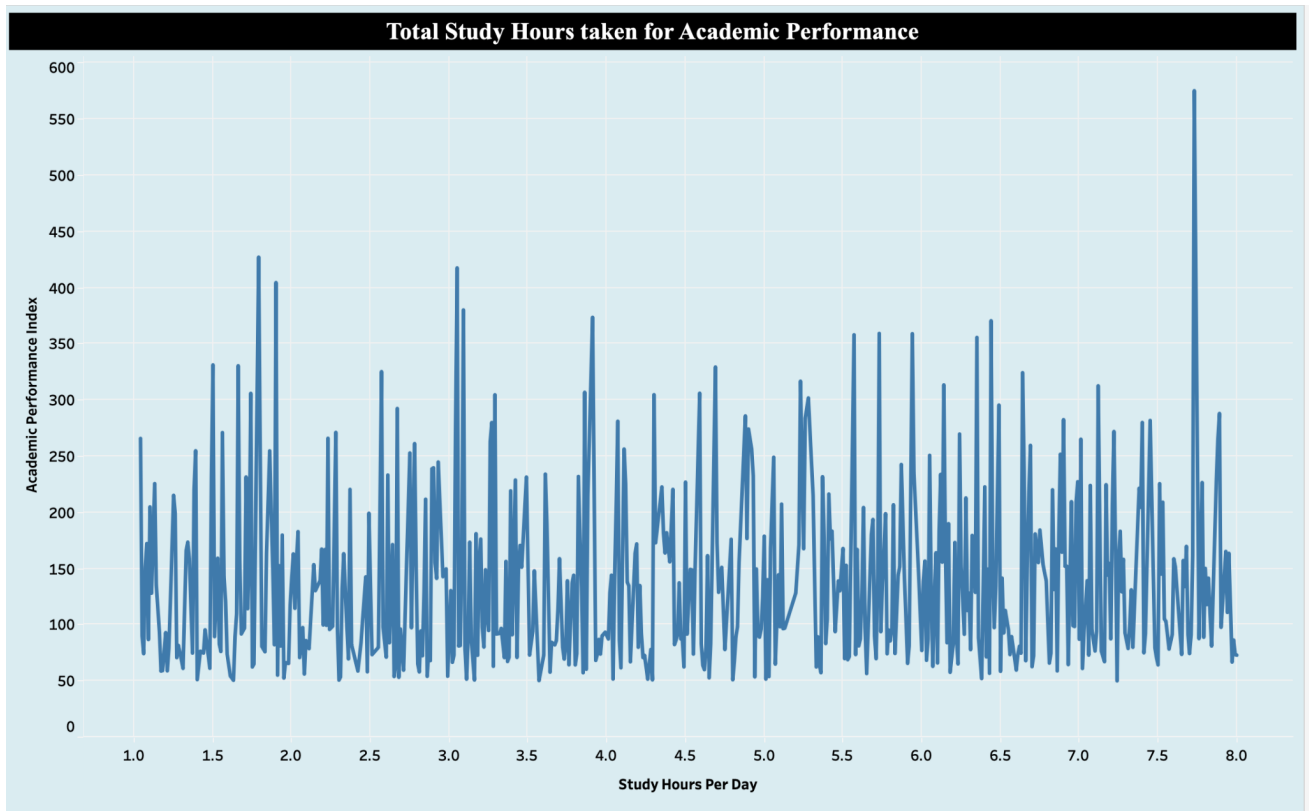


Figure 5: Study Hours Per Day vs Academic Performance Index

9.5 Gender vs Depression Level

The pie chart displays the sum of the recorded Depression Level by gender. The displayed totals are 1,818 for Female, 1,825 for Male and 1,860 for Other, giving a total of 5,503. The three values are relatively close, with Other having the highest total in this aggregation.



Figure 6: Gender vs Depression Level

10. Key Dashboard Findings

- The dataset contains 1,000 student records.
- The dashboard shows an average stress level of 5.452.
- The displayed average sleep value is 6.487.
- Mental health risk-related record counts are 331 Female, 341 Male and 328 Other.
- The gender-wise mental health risk counts are very close.
- The gender-wise heart-rate-variability observation counts are also close.
- The depression-level total shown is 5,503, split across Female (1,818), Male (1,825) and Other (1,860).
- The study-hours/performance chart shows considerable variation, so increasing study hours alone does not give a uniform academic-performance outcome in the displayed data.

11. Interpretation of the Results

The dashboard brings academic and well-being indicators together. The sample does not show a dramatic difference in the number of records across gender categories, making comparisons easier.

The stress-related view indicates that the average values across gender are in a fairly narrow range. Similarly, the depression totals are close when viewed using the selected SUM measure. These results should therefore be discussed carefully rather than treated as major gender-based differences.

The study-hours chart shows how varied student outcomes can be. Students with similar study hours do not necessarily have the same academic performance index, suggesting that academic performance can be affected by several factors.

12. Limitations

- The analysis is based only on the variables available in the provided dataset.
- The dashboard is descriptive and does not establish cause-and-effect relationships.
- Some charts use aggregated values, so individual student-level differences are not visible in the summaries.
- The HRV chart uses a count of observations and should not be read as an average physiological value.
- Mental health indicators in a dataset cannot replace professional psychological or medical assessment.
- The project does not include qualitative information such as interviews or student feedback.

13. Future Scope

- Add filters for age, academic year, course or other available student characteristics.
- Use scatter plots and trend lines to examine relationships more directly.
- Include correlation analysis between stress, sleep, study hours, depression and academic performance.
- Add time-based data if repeated observations are available.
- Develop a privacy-conscious risk-monitoring view for student support.
- Improve the dashboard with interactive tooltips and drill-down views.

14. Conclusion

This project demonstrates how Tableau can be used to turn a student dataset into a practical visual report. The final dashboard gives a compact view of stress, sleep, mental health risk, depression, study hours, academic performance and heart-rate-variability-related information.

The analysis shows that the student mental-health ecosystem is not represented by a single variable. Different indicators need to be considered together, and visual analysis makes it easier to compare these indicators and notice patterns. Overall, the project provided useful hands-on experience in data preparation, visualization, dashboard creation and communicating analytical findings.

15. References / Sources

- Student dataset provided for the academic project.
- Tableau – data visualization and dashboard development software.
- Tableau official documentation and learning resources.